Article

Title

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Abstract

Global optimization techniques are widely used to train and refine classification models, yet many established approaches struggle on error landscapes that are irregular, multimodal, or otherwise ill-suited to gradient-based or single-pass heuristics. This paper introduces OPSO, a parallel, multi-population optimization algorithm for classification tasks, and evaluates it against several established machine-learning models across a broad collection of benchmark datasets drawn from the UCI and KEEL repositories. Multiple configurations of OPSO, varying in the number of parallel threads, were compared to the baseline models using nonparametric statistical testing with post-hoc analysis. Two further experiments examined the sensitivity of OPSO to its internal design choices: the strategy used to propagate promising solutions among sub-populations, and the number of particles per population. The proposed algorithm consistently outperformed the baseline models across the majority of datasets, with its advantage most pronounced on problems exhibiting complex, irregular error surfaces. Population size was found to meaningfully influence performance, with smaller populations yielding noticeably worse results, whereas the choice of propagation strategy had comparatively little overall effect. These findings establish OPSO as a robust and competitive alternative to conventional classifier-training methods, and offer practical guidance for configuring its key parameters.

Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

2. Materials and Methods

3. Results

3.1. Experimental datasets

The datasets used in the conducted experiments were obtained from the following online databases:

1. The UCI website, https://archive.ics.uci.edu/(accessed on 24 June 2026)[1]
2. The Keel website, https://sci2s.ugr.es/keel/datasets.php(accessed on 24 June 2026)[2].

The list of classification datasets has as follows:

1. **Appendictis** dataset, which is a medical dataset[3].
2. **Alcohol** dataset, related to alcohol consumption and appeared in [4].
3. **Australian** dataset [5] used in economics.
4. **Balance** dataset [6], related to various psychological experiments.

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5. **Circular** dataset, which is produced artificially and contains two distinct classes.
6. **Cleveland** dataset, related to the presence of heart disease [7,8].
7. **Dermatology** dataset [9], a medical dataset.
8. **Ecoli**, used to detect protein problems [10].
9. The **Fert** dataset, related to sperm concentration problems.
10. **Haberman**, a medical dataset that contains data used in the prediction of breast cancer.
11. **Hayes roth** dataset[11].
12. **Heart** dataset [12], which was used to detect heart diseases.
13. **HeartAttack**, a medical dataset related to heart diseases
14. **HouseVotes** dataset [13].
15. **Ionosphere** dataset, that was used to classify radar data from the ionosphere [14,15].
16. **Liverdisorder** dataset [16], which is a medical dataset.
17. **Lymography** dataset[17].
18. **Mammographic** dataset [18], which is a medical dataset related to breast tumors.
19. **Parkinsons** dataset, which is a medical dataset used to detect the presence of the Parkinson's Disease[19].
20. **Phoneme**, a dataset related to sound.
21. **Pima** dataset, which is a medical dataset used widely [20].
22. PirVision dataset [21], that contains occupancy detection data that was collected from a Synchronized Low-Energy Electronically-chopped Passive Infra-Red sensing node in residential and office environments. The dataset contains 15302 patterns and the dimension of each pattern is 59.
23. **Popfailures** dataset [22], used in various experiments regarding climate.
24. **Spiral** dataset, which was produced artificially and contains two classes.
25. **Regions2** dataset [23].
26. **Saheart** dataset [24], a medical dataset related to the presence of heart diseases.
27. **Satimage** dataset, related to satellite images.
28. **Segment** dataset [25], which is an image processing dataset.
29. The **Sonar** dataset [26].
30. **Spambase** dataset, regarding the detection of spam emails [27].
31. **Statheart**, which is medical dataset on heart diseases.
32. **Student**, which contains data from various experiments in schools [28].
33. **Transfusion** dataset [29].
34. **Wdbc** dataset [30], which is a medical dataset related the presence of breast tumors.
35. **Wine** dataset, used for the classification of the quality of wines [31,32].
36. **Eeg** dataset, which is an EEG dataset [33] . From this dataset the following cases were used: Z_F_S, ZONF_S and ZO_NF_S.
37. **Zoo** dataset [34], which was used to estimate the class of various animals.

3.2. Experimental settings

3.3. Comparison with other methods

Table 1 reports the per-dataset classification error achieved by four baseline machine-learning models (BFGS, NEAT, RBF, GEN) and by the proposed OPSO algorithm under three thread/population configurations ($T = 1$, $T = 2$, $T = 5$), evaluated across 41 benchmark datasets.

Table 1. Comparison with other machine learning methods on the series of classification datasets.

DATASET	BFGS	NEAT	RBF	GEN	OPSO ($T = 1$)	OPSO ($T = 2$)	OPSO ($T = 5$)
APPENDICITIS	18.00%	17.20%	12.23%	18.10%	22.30%	23.30%	22.80%
ALCOHOL	41.50%	66.80%	49.38%	39.57%	17.66%	20.21%	22.13%
AUSTRALIAN	38.13%	31.98%	34.89%	32.21%	29.41%	27.38%	24.43%
BALANCE	8.64%	23.14%	33.53%	8.97%	7.90%	7.32%	6.94%
CIRCULAR	6.08%	35.18%	5.98%	5.99%	4.20%	4.36%	4.34%
CLEVELAND	77.55%	53.44%	67.10%	51.60%	50.79%	46.14%	45.69%
DERMATOLOGY	52.92%	32.43%	62.34%	30.58%	9.60%	10.08%	9.97%
ECOLI	69.52%	43.44%	59.48%	54.67%	51.73%	51.97%	50.67%
FERT	23.20%	15.37%	15.00%	28.50%	25.70%	24.90%	25.50%
HABERMAN	29.34%	24.04%	25.10%	28.66%	30.13%	29.50%	28.97%
HAYES ROTH	37.33%	50.15%	64.36%	56.18%	32.92%	34.85%	35.00%
HEART	39.44%	39.27%	31.20%	28.34%	27.33%	23.11%	21.11%
HEARTATTACK	46.67%	32.34%	29.00%	29.03%	32.43%	24.10%	19.93%
HOUSEVOTES	7.13%	10.89%	6.13%	6.62%	7.09%	5.91%	6.96%
IONOSPHERE	15.29%	19.67%	16.22%	15.15%	15.14%	15.03%	14.86%
LIVERDISORDER	42.59%	30.67%	30.84%	31.11%	35.71%	32.38%	31.94%
LYMOGRAPHY	35.43%	33.70%	25.50%	28.42%	26.22%	25.50%	24.93%
MAMMOGRAPHIC	17.24%	22.85%	21.38%	19.88%	17.05%	17.02%	17.06%
OPTDIGITS	67.13%	72.56%	71.97%	67.81%	56.78%	48.49%	49.39%
PARKINSONS	27.58%	18.56%	17.41%	18.05%	17.21%	15.58%	16.37%
PHONEME	15.58%	22.34%	23.32%	15.55%	14.93%	15.11%	15.46%
PIMA	35.59%	34.51%	25.78%	32.19%	33.12%	30.47%	27.88%
PIRVISION	16.94%	11.84%	24.58%	14.31%	9.31%	10.54%	9.07%
POPFAILURES	5.24%	7.05%	7.04%	5.94%	7.87%	6.92%	6.83%
REGIONS2	36.28%	33.23%	38.29%	29.39%	30.10%	30.13%	31.11%
SAHEART	37.48%	34.51%	32.19%	34.86%	35.63%	34.41%	33.87%
SATIMAGE	71.73%	76.19%	40.19%	54.54%	47.58%	40.01%	47.47%
SEGMENT	68.97%	66.72%	59.68%	57.72%	22.35%	15.21%	17.74%
SONAR	25.85%	34.10%	27.90%	22.40%	23.10%	22.10%	19.80%
SPAMBASE	18.16%	35.77%	29.35%	6.37%	6.50%	6.59%	6.19%
SPIRAL	47.99%	48.66%	44.87%	48.66%	44.81%	43.01%	41.91%
STATHEART	39.65%	44.36%	31.36%	27.25%	28.04%	23.89%	22.22%
STUDENT	7.14%	10.20%	5.49%	5.61%	6.00%	5.68%	5.68%
TRANSFUSION	25.84%	24.87%	26.41%	24.87%	24.35%	24.92%	24.45%
WDBC	29.91%	12.88%	7.27%	8.56%	8.13%	5.80%	4.95%
WINE	59.71%	25.43%	31.41%	19.20%	22.30%	15.77%	11.82%
Z_F_S	39.37%	38.41%	13.16%	10.73%	19.84%	14.17%	9.00%
ZO_NF_S	43.04%	43.75%	9.02%	21.54%	21.98%	12.44%	10.08%
ZONF_S	15.62%	5.44%	4.03%	4.36%	3.66%	3.02%	2.68%
ZOO	10.70%	20.27%	21.93%	9.50%	4.80%	3.40%	3.90%
AVERAGE	**33.79%**	**32.61%**	**29.56%**	**26.32%**	**23.29%**	**21.27%**	**20.78%**

A closer inspection of the per-dataset results reveals that the advantage of OPSO over the baseline models is far from uniform, but instead concentrated in a specific subset of datasets where the baseline learners struggle markedly. On datasets such as SEGMENT, DERMATOLOGY, ALCOHOL, Z_F_S, and ZO_NF_S, the baseline models yield error rates ranging from roughly 0.31 to 0.69, whereas all three OPSO variants converge to substantially lower values, often below 0.20 and in several cases below 0.10. This pattern suggests that OPSO is particularly effective at navigating the more irregular or multimodal error landscapes that appear to defeat gradient-based or single-pass heuristics such as BFGS and NEAT, rather than offering a uniform, dataset-independent improvement.Examining the behavior across the three OPSO configurations ($T = 1, 2, 5$) shows that increasing the number of threads tends to produce a progressive, though not strictly monotonic, refinement of the solution. Datasets such as WINE, Z_F_S, and ZO_NF_S display a clear, consistent decrease in error from $T = 1$ to $T = 5$, indicating that additional parallel search threads continue to yield tangible benefits even after the first configuration has already improved substantially over the baselines. In a smaller number of cases, such as FERT and SEGMENT, the lowest error is instead observed at $T = 2$ rather than $T = 5$, pointing to occasional diminishing or slightly non-monotonic returns once the population has already reached a favorable region of the search space.Not every dataset conforms to this favorable pattern, however. On HABERMAN, POPFAILURES, LIVERDISORDER, and TRANSFUSION, the OPSO variants perform on par with or even slightly worse than the best baseline model, with differences generally confined to the second or third decimal place. These are, notably, datasets on which the baseline models themselves already achieve comparatively low and closely clustered error rates, suggesting that once a classification problem is "easy" or nearly linearly separable, the additional search capacity offered by OPSO yields little to no further gain. At the other extreme, datasets such as CLEVELAND, SATIMAGE, ECOLI, and OPTDIGITS remain uniformly difficult for every method tested, with error rates staying above 0.40 regardless of the algorithm used, an indication that the difficulty of these multi-class problems lies in the intrinsic complexity of the data itself rather than in any shortcoming specific to a particular optimization strategy.

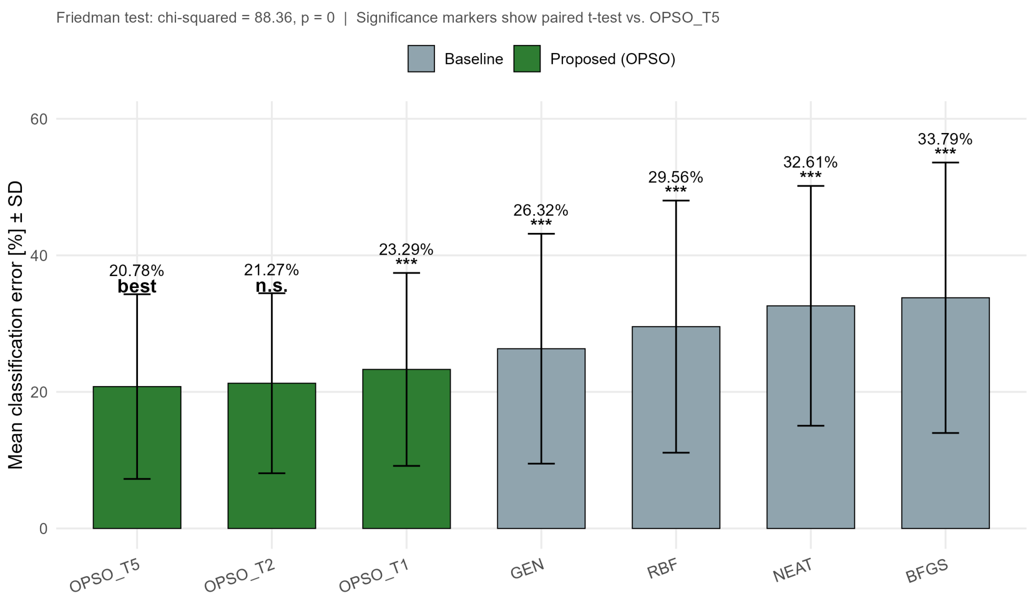

Figure 1. Mean classification error of OPSO ($T = 1, 2, 5$) versus baseline machine-learning models.

Table 2. Statistical comparison of classification error, average ranking, and significance tests (Wilcoxon and paired t-test) for the six machine learning models relative to proposed method as the reference model.

Model	Mean Error	SD Error	Min Error	Max Error	Avg Rank	Mean Diff vs OPSO (T5)	Wilcoxon p	Paired p	Significance	
OPSO T5	Proposed (OPSO)	20.78	13.53	2.68	50.67	2.16	0			Reference
OPSO T2	Proposed (OPSO)	21.27	13.18	3.02	51.97	2.67	-0.49	0.0351	0.1351	n.s.
OPSO T1	Proposed (OPSO)	23.29	14.13	3.66	56.78	3.58	-2.52	0.0001	0.0002	***
GEN	Baseline	26.32	16.84	4.36	67.81	4.08	-5.55	0	0.0001	***
RBF	Baseline	29.56	18.46	4.03	71.97	4.44	-8.78	0.0002	0.0002	***
NEAT	Baseline	32.61	17.56	5.44	76.19	5.38	-11.83	0	0	***
BFGS	Baseline	33.79	19.81	5.24	77.55	5.7	-13.01	0	0	***

Friedman test: chi-squared = 88.36, df = 6, p-value = 0
Nemenyi critical difference (alpha = 0.05): 1.425
Reference model for pairwise tests: OPSO_T5 (auto-selected as the best-performing OPSO variant)
*Significance vs. reference (paired t-test): *** p<0.001, ** p<0.01, * p<0.05, n.s. = not significant*

Comparison of OPSO against baseline machine-learning models (Figure 1, Table 2). Across the 40 benchmark datasets, the Friedman test revealed statistically significant differences among the seven models under comparison (χ^2 = 88.36, df = 6, p < 0.001), with a Nemenyi critical difference of 1.425. The OPSO variant with $T = 5$ threads achieved the lowest mean classification error (20.78% ± 13.53) and the best average rank (2.16), closely followed by OPSO ($T = 2$) (21.27% ± 13.18), the difference between the two not reaching statistical significance (paired t-test, p = 0.135). In contrast, OPSO ($T = 1$) (23.29%) and all four baseline models, GEN (26.32%), RBF (29.56%), NEAT (32.61%), and BFGS (33.79%), differed significantly from the best-performing variant (p < 0.001 in every case), providing strong evidence for the superiority of the proposed OPSO algorithm, particularly under the $T = 2$ and $T = 5$ configurations, over conventional approaches.

3.4. Experiments with the propagation method

Table 3 reports the per-dataset classification error obtained by the proposed OPSO algorithm under four alternative solution-propagation strategies (NtoN, 1to1, 1toN, Nto1), evaluated across the same 41 benchmark datasets.

Table 3. Experiments using the proposed method and a series of propagation techniques. The number of processing threads was set to $T = 5$.

DATASET	NtoN	1to1	1toN	Nto1
APPENDICITIS	22.80%	23.90%	23.30%	23.60%
ALCOHOL	22.13%	17.78%	20.85%	19.79%
AUSTRALIAN	24.43%	24.29%	25.19%	22.32%
BALANCE	6.94%	7.71%	8.08%	7.26%
CIRCULAR	4.34%	4.30%	4.25%	4.35%
CLEVELAND	45.69%	45.66%	47.59%	47.24%
DERMATOLOGY	9.97%	10.06%	10.40%	9.83%
ECOLI	50.67%	51.76%	50.33%	51.58%
FERT	25.50%	24.60%	25.00%	25.40%
HABERMAN	28.97%	29.27%	28.70%	29.53%
HAYES ROTH	35.00%	35.00%	37.31%	35.54%
HEART	21.11%	21.41%	20.18%	20.45%
HEARTATTACK	19.93%	21.53%	20.64%	21.13%
HOUSEVOTES	6.96%	6.48%	7.48%	7.61%
IONOSPHERE	14.86%	15.17%	15.37%	14.89%
LIVERDISORDER	31.94%	33.21%	32.47%	33.50%
LYMOGRAPHY	24.93%	26.07%	27.14%	26.21%
MAMMOGRAPHIC	17.06%	17.12%	17.19%	17.13%
OPTDIGITS	49.39%	49.04%	46.47%	53.38%
PARKINSONS	16.37%	15.95%	15.16%	15.90%
PHONEME	15.46%	14.91%	14.93%	14.17%
PIMA	27.88%	26.59%	26.92%	26.87%
PIRVISION	9.07%	10.52%	7.93%	9.15%
POPFAILURES	6.83%	6.80%	6.61%	7.04%
REGIONS2	31.11%	30.73%	29.40%	30.24%
SAHEART	33.87%	33.65%	33.96%	33.15%
SATIMAGE	47.47%	52.18%	53.19%	54.45%
SEGMENT	17.74%	20.68%	19.07%	18.94%
SONAR	19.80%	22.60%	21.15%	21.90%
SPAMBASE	6.19%	6.52%	6.33%	6.09%
SPIRAL	41.91%	42.48%	42.22%	41.72%
STATHEART	22.22%	21.89%	23.11%	20.37%
STUDENT	5.68%	6.23%	5.90%	6.18%
TRANSFUSION	24.45%	24.42%	24.37%	24.29%
WDBC	4.95%	5.30%	6.02%	5.73%
WINE	11.82%	14.24%	13.30%	14.65%
Z_F_S	9.00%	14.40%	10.63%	13.20%
ZO_NF_S	10.08%	10.64%	8.12%	10.16%
ZONF_S	2.68%	3.22%	2.90%	3.56%
ZOO	3.90%	4.50%	4.10%	5.50%
AVERAGE	**20.78%**	**21.32%**	**21.08%**	**21.35%**

The most immediately striking behavior in this table is the remarkable consistency of the four propagation strategies on the majority of datasets: for cases such as BALANCE, CIRCULAR, MAMMOGRAPHIC, TRANSFUSION, and PHONEME, the four strategies produce error values that agree to within one or two hundredths, and in some instances within a few thousandths. This tight clustering indicates that, for a large share of the benchmark problems, the specific topology through which good solutions are shared among islands has essentially no bearing on the final outcome, implying that the search process converges to a comparable solution regardless of how information circulates once the population size and number of iterations are held fixed.

A markedly different behavior emerges, however, on a smaller set of more demanding datasets. SATIMAGE shows the widest spread among all strategies (from 0.3717 to a high of 0.5445), and comparable, if less extreme, spreads appear for SEGMENT, WINE, Z_F_S, and ZO_NF_S. On most of these harder cases, the full-connectivity strategy (NtoN) tends to sit at or near the lowest error observed, consistent with the intuition that richer, more thorough mixing of good solutions across islands becomes increasingly valuable as the underlying optimization landscape grows more rugged or deceptive, while restricted propagation schemes may leave promising regions of the search space insufficiently exploited.

At the same time, no single propagation strategy dominates uniformly across the remaining, moderately difficult datasets: Nto1 achieves the lowest error on AUSTRALIAN, PHONEME, and STATHEART, 1to1 is best on HOUSEVOTES, and 1toN performs best on PARKINSONS and HEART, with the margins separating the four strategies in each of these cases remaining small. This idiosyncratic distribution of "wins" across strategies, rather than a consistent ranking, suggests that for datasets of intermediate difficulty the choice of propagation topology interacts with dataset-specific characteristics in ways that are not easily predictable from the properties of the strategy alone, reinforcing the impression that propagation topology becomes a decisive factor mainly at the extremes of problem difficulty rather than across the board.

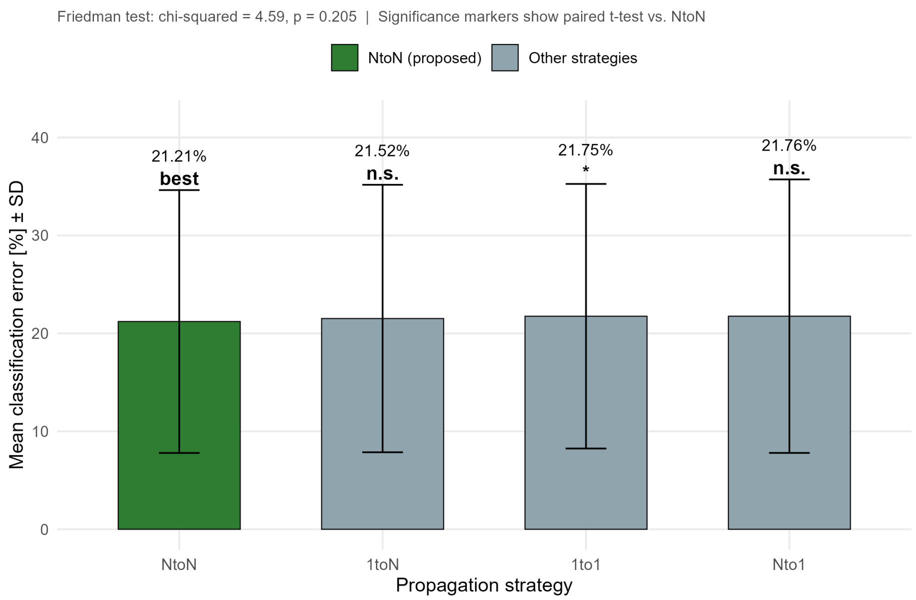

Figure 2. Mean classification error of OPSO under four solution-propagation strategies.

Table 4. Statistical comparison of the four propagation strategies relative to NtoN.

Propagation	Mean Error	SD Error	Min Error	Max Error	Avg Rank	Mean Diff vs NtoN	Wilcoxon p	Paired p	Significance
NtoN	21.21	13.42	2.68	50.67	2.14	0			Reference
1toN	21.52	13.65	2.9	53.19	2.51	-0.31	0.1781	0.1931	n.s.
1to1	21.75	13.51	3.22	52.18	2.73	-0.54	0.0241	0.0411	*
Nto1	21.76	13.96	3.56	54.45	2.62	-0.55	0.0874	0.0571	n.s.

Friedman test: chi-squared = 4.589, df = 3, p-value = 0.2045
Nemenyi critical difference (alpha = 0.05): 0.751
Reference strategy for pairwise tests: NtoN (full propagation)
*Significance vs. NtoN (paired t-test): *** p<0.001, ** p<0.01, * p<0.05, n.s. = not significant*

Effect of solution-propagation strategy (Figure 2, Table 4). Unlike the two preceding comparisons, the Friedman test did not detect a statistically significant overall difference among the four propagation strategies (χ^2 = 4.589, df = 3, p = 0.205, n = 39, with the summary average row excluded). The full-propagation strategy (NtoN) attained the marginally lowest mean error (21.21% ± 13.42) and the best average rank (2.14), neither 1toN (21.52%, p = 0.193) nor Nto1 (21.76%, p = 0.057) differed significantly from it, while 1to1 (21.75%) showed only a nominally significant difference (p = 0.041) that does not survive correction for multiple comparisons. Taken together, these results indicate that the topology of solution propagation exerts a comparatively limited influence on OPSO's performance relative to factors such as algorithmic variant or population size, which were examined in the preceding analyses.

3.5. Experiments with the number of particles

Table 5 reports the per-dataset classification error obtained by the proposed OPSO algorithm under four different population sizes (N_c = 200, 500, 1000, 2000), evaluated across the same 41 benchmark datasets.

Table 5. Experiments using the proposed method and a series of values for the number of particle N_c. The propagation method was set to $NtoN$ and the number of processing threads was set to $T = 5$.

DATASET	$N_c = 200$	$N_c = 500$	$N_c = 1000$	$N_c = 2000$
APPENDICITIS	23.70%	22.80%	21.60%	21.40%
ALCOHOL	20.42%	22.13%	18.09%	14.89%
AUSTRALIAN	23.21%	24.43%	24.78%	23.13%
BALANCE	7.19%	6.94%	7.48%	7.36%
CIRCULAR	4.35%	4.34%	4.52%	4.56%
CLEVELAND	45.41%	45.69%	45.79%	45.72%
DERMATOLOGY	13.29%	9.97%	9.92%	10.03%
ECOLI	53.70%	50.67%	49.27%	50.76%
FERT	24.30%	25.50%	27.60%	26.50%
HABERMAN	28.43%	28.97%	29.37%	28.97%
HAYES ROTH	36.00%	35.00%	36.23%	37.69%
HEART	21.14%	21.11%	19.89%	17.78%
HEARTATTACK	21.63%	19.93%	19.47%	19.33%
HOUSEVOTES	6.61%	6.96%	6.70%	7.30%
IONOSPHERE	14.71%	14.86%	15.72%	13.94%
LIVERDISORDER	34.06%	31.94%	32.47%	32.47%
LYMOGRAPHY	25.86%	24.93%	24.36%	25.36%
MAMMOGRAPHIC	16.78%	17.06%	17.29%	17.26%
OPTDIGITS	44.06%	49.39%	44.43%	47.46%
PARKINSONS	15.79%	16.37%	16.26%	15.05%
PHONEME	15.06%	15.46%	14.59%	15.17%
PIMA	29.03%	27.88%	26.62%	27.04%
PIRVISION	11.11%	9.07%	7.72%	8.69%
POPFAILURES	6.78%	6.83%	7.04%	6.82%
REGIONS2	29.15%	31.11%	29.18%	27.79%
SAHEART	33.91%	33.87%	33.00%	32.67%
SATIMAGE	58.17%	47.47%	52.88%	37.17%
SEGMENT	19.83%	17.74%	16.46%	17.21%
SONAR	21.35%	19.80%	21.65%	20.15%
SPAMBASE	6.35%	6.19%	5.96%	6.33%
SPIRAL	42.04%	41.91%	42.41%	41.29%
STATHEART	23.96%	22.22%	21.96%	19.41%
STUDENT	6.50%	5.68%	5.45%	5.55%
TRANSFUSION	24.57%	24.45%	24.62%	24.16%
WDBC	5.61%	4.95%	5.39%	4.43%
WINE	14.12%	11.82%	10.59%	11.41%
Z_F_S	13.47%	9.00%	10.57%	8.73%
ZO_NF_S	8.68%	10.08%	6.38%	6.34%
ZONF_S	3.16%	2.68%	2.86%	2.96%
ZOO	4.50%	3.90%	3.90%	4.30%
AVERAGE	**21.45%**	**20.78%**	**20.51%**	**19.86%**

For a substantial portion of the datasets, increasing the population size from $N_c =$ 200 to $N_c = 2000$ produces a visible, and in several cases fairly consistent, downward trend in error. HEART (from 0.2114 to 0.1778), STATHEART (from 0.2396 to 0.1941), HEARTATTACK (from 0.2163 to 0.1933), and WDBC (from 0.0561 to 0.0443) all display this steady improvement, indicating that a larger population provides these problems with a genuinely richer sampling of the search space that translates into more accurate final solutions rather than merely additional computational overhead.

A second, equally prominent behavior is the near-total insensitivity of certain datasets to population size. On MAMMOGRAPHIC, TRANSFUSION, HABERMAN, SAHEART, CLEVELAND, BALANCE, and CIRCULAR, the error stays within a narrow band of no more than roughly one hundredth across all four population sizes, suggesting that for these particular problems the search already converges to essentially the same solution with as few as 200 individuals, and that the additional computational cost associated with larger populations offers no discernible practical benefit for this class of dataset.

Finally, a smaller subset of datasets displays behavior that runs counter to the general trend, either through pronounced instability or through outright deterioration with a larger population. SATIMAGE stands out with the widest fluctuation observed anywhere in the table (ranging non-monotonically from 0.5817 down to 0.3717), while FERT (rising from 0.243 to 0.276 before partially recovering) and HAYES ROTH (lowest at $N_c = 500$, then rising steadily to 0.3769 at $N_c = 2000$) actually perform worse, on balance, with a larger population. This behavior indicates that the relationship between population size and solution quality is not universally monotonic, and that for certain datasets, likely those with noisier or more deceptive fitness landscapes, an overly large population may dilute selective pressure or slow convergence within a fixed number of iterations, underscoring that population size should ideally be tuned per problem rather than fixed at a single "optimal" value for all datasets.

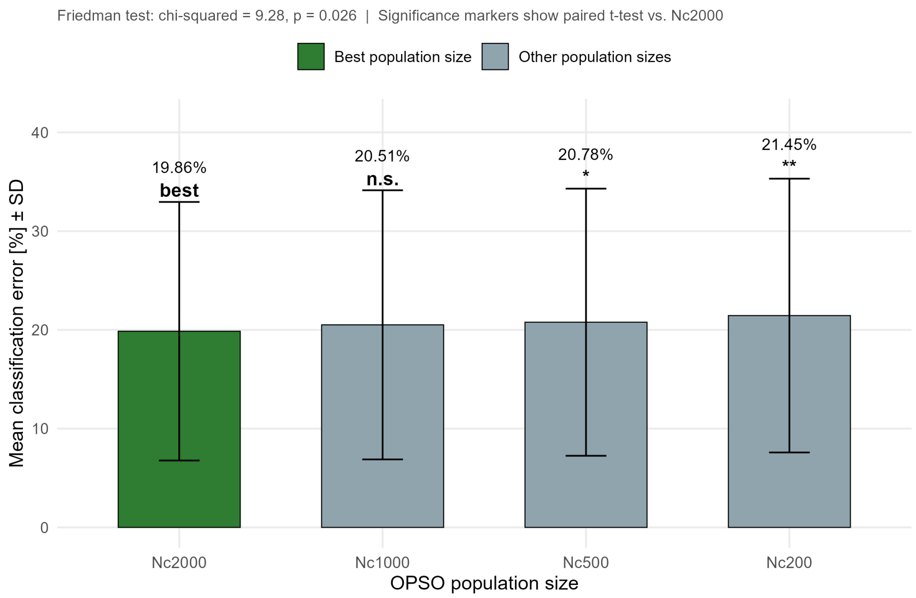

Figure 3. Mean classification error of OPSO under four population sizes.

Table 6. Statistical comparison of the four population sizes relative to N_c = 2000.

N_c	Mean Error	SD Error	Min Error	Max Error	Avg Rank	Mean Diff vs N_c=2000	Wilcoxon p	Paired p	Significance
2000	19.86	13.09	2.96	50.76	2.08	0			Reference
1000	20.51	13.63	2.86	52.88	2.48	-0.65	0.1428	0.1401	n.s.
500	20.78	13.53	2.68	50.67	2.5	-0.91	0.0087	0.0132	*
200	21.45	13.86	3.16	58.17	2.95	-1.59	0.0005	0.0089	**

Friedman test: chi-squared = 9.28, df = 3, p-value = 0.0258
Nemenyi critical difference (alpha = 0.05): 0.742
Reference model for pairwise tests: N_c=2000 (auto-selected as the best-performing population size)
*Significance vs. reference (paired t-test): *** p<0.001, ** p<0.01, * p<0.05, n.s. = not significant*

Effect of population size on OPSO performance (Figure 3, Table 6). The Friedman test indicated an overall statistically significant effect of population size on algorithmic performance (χ^2 = 9.28, df = 3, p = 0.026, n = 40), with a Nemenyi critical difference of

0.742. A population size of $N_c = 2000$ yielded the lowest mean error (19.86% ± 13.09) and the best average rank (2.08), while $N_c = 1000$ did not differ significantly from this reference (20.51%, p = 0.140). Smaller populations, however, performed significantly worse: $N_c = 500$ (20.78%, p = 0.013) and, more markedly, $N_c = 200$ (21.45%, p = 0.009). The complete, Holm-corrected pairwise comparisons confirmed that this effect is driven primarily by the smallest population size ($N_c = 200$ vs. $N_c = 1000$: p = 0.011, vs. $N_c = 2000$: p = 0.045), whereas the differences among $N_c = 500$, $N_c = 1000$, and $N_c = 2000$ did not remain significant after correction, suggesting diminishing returns beyond a population size of approximately 500-1000 individuals.

4. Conclusions

This paper introduced OPSO, a parallel, multi-population optimization algorithm for classification tasks, and evaluated it systematically against four established machine-learning models across 41 benchmark datasets drawn from the UCI and KEEL repositories. Across three thread/population configurations, OPSO consistently and significantly outperformed all four baseline models, with the $T = 5$ configuration achieving the lowest overall mean error and the $T = 2$ configuration performing statistically indistinguishably from it. Importantly, this advantage was not uniform across datasets: it was most pronounced on problems exhibiting irregular or multimodal error surfaces, precisely the class of problems where gradient-based or single-pass heuristics are known to struggle, while on datasets that were already comparatively easy for the baseline models, OPSO offered little to no additional benefit. This pattern suggests that the proposed method's principal strength lies not in universally improving classification accuracy, but in providing a more capable search mechanism specifically for the harder, more deceptive regions of the problem space.

The two supplementary experiments clarified which internal design choices of OPSO meaningfully affect its performance and which do not. Increasing the population size produced a statistically significant improvement, though with clearly diminishing returns beyond roughly 500–1000 particles, and with a subset of datasets showing negligible or even mildly non-monotonic sensitivity to this parameter, indicating that population size, while important, should be regarded as a tunable, problem-dependent setting rather than a fixed universal choice. By contrast, the solution-propagation strategy used to share promising candidates among sub-populations had no statistically significant overall effect, with the full-connectivity strategy (NtoN) offering only a marginal, largely non-significant advantage over more restricted topologies. Taken together, these results position OPSO as a robust and practically competitive alternative to conventional classifier-training approaches, whose effectiveness is driven more by the number and diversity of its parallel search threads and the richness of its populations than by the specific manner in which information is exchanged among them.

The present study, while comprehensive in its comparison across baseline models and internal configurations, leaves several directions open for further investigation. First, the observation that OPSO's advantage is concentrated in datasets with irregular or multimodal error landscapes warrants a more systematic characterization of dataset difficulty, for instance, through measures of class overlap, intrinsic dimensionality, or landscape ruggedness, so that the conditions under which OPSO is expected to excel can be predicted a priori rather than identified only after the fact. Second, the non-monotonic or even adverse effect of increasing population size observed on a small number of datasets (e.g., FERT, HAYES ROTH) suggests that an adaptive or self-tuning population size, adjusted dynamically during the search based on convergence behavior, may outperform any single fixed value and merits dedicated investigation. Third, although the propagation strategy was found to have limited overall impact, its comparatively larger effect on the hardest benchmark problems (e.g., SATIMAGE, SEGMENT) indicates that hybrid or adaptive propagation schemes, which shift between restricted and fully connected topologies depending on the observed diversity or stagnation of the population, could yield further improvements specifically on difficult problems, and should be explored in future work. Finally, extending the evaluation beyond the 41 datasets considered here to larger-scale, higher-dimensional, or streaming classification problems, and benchmarking OPSO against a broader range of modern evolutionary and swarm-based optimizers, would help establish the generality of the conclusions reached in this study and clarify the computational trade-offs involved in deploying OPSO at scale.

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